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When Valid Counterfactuals Fail as Actions: An Evidence-Based Evaluation of Causal Actionable Recourse

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Computational results are conditional on the specified structural models and action constraints. No human participant study is reported.

ABSTRACT A counterfactual state that changes a classifier's decision need not describe an action whose consequences preserve that decision. We study this distinction using Causal Actionable Counterfactuals, an implementation that searches over interventions and evaluates the states produced by a structural causal model. The evaluation separates proposed-state validity, replayed-action success, action feasibility, and state fidelity. A plan is usable under the specified model only when the same suggestion passes replay and feasibility. We audit saved experiments on Adult, German Credit, HELOC, and diabetes, comprising 294 rejected applicants in the primary comparison. At the canonical search seed, the intervention method provides usable recourse for every queried applicant, while several baselines tie it on three datasets. On HELOC, usable rates range from 18.0% to 100.0%; proposed-state validity alone obscures much of this gap. However, saved search repetitions expose incomplete seed coverage and generation failures: the intervention method's mean usable rate on diabetes is 94.4%. Recomputed applicant-level inference avoids treating repeated seeds as independent people. A refitted-model sensitivity analysis and two synthetic worlds further delimit the shared-model findings. These results support reporting replay success and feasibility alongside predictive validity, while providing no guarantee of real-world intervention effects or universal method superiority.

INDEX TERMS Algorithmic recourse, black-box models, causal inference, counterfactual explanations, reproducibility, structural causal models.

I. INTRODUCTION

ALGORITHMIC recourse asks what a person can change after receiving an unfavorable prediction. A common computational answer is a nearby feature vector that receives the desired label. This answer has a precise predictive meaning: the classifier accepts that vector. Its interpretation as advice requires additional assumptions about which coordinates a person controls and how the rest of the record responds.

For example, an applicant may change education while occupation and hours subsequently respond. Directly editing all three coordinates describes a different action from intervening on education alone. A method that reports a resulting state therefore needs to distinguish its requested interventions from their predicted consequences. The distinction between feature changes and interventions is established in causal recourse research [4]. Our work uses that distinction to organize an empirical audit of an existing implementation and its saved experiments.

The implementation, Causal Actionable Counterfactuals (CAC), uses a population search over a bounded intervention set. Each candidate passes through a fitted structural causal model (SCM) before its classifier score is evaluated. The resulting search space is internally consistent with that model, subject to the implementation's encoding and rounding. Internal consistency alone does not establish that a favorable outcome is reachable, that the search will find it, or that the assumed causal equations describe the world.

This paper asks three questions. First, how much does an evaluation based only on proposed-state validity conceal about replay success and action constraints? Second, what changes when causal propagation and objective terms are removed from the same search? Third, how do search repetitions, alternative fitted equations, and known synthetic mechanisms limit the primary result?

The contribution is an evidence-based evaluation framework and a verified account of this implementation. We define all endpoints at the suggestion and applicant levels, reconstruct the primary table from proposal-level artifacts,

and replay the saved plans against the stored models. We also recompute paired inference using applicants as independent units, retaining incomplete runs and recorded failures. The findings establish useful computational distinctions; they do not introduce intervention-based recourse as a new concept.

II. RELATED WORK

A. COUNTERFACTUAL EXPLANATIONS AND FEASIBLE CHANGES

Wachter *et al.* [1] formulate counterfactual explanations through alternative inputs that receive a desired decision. DiCE promotes diverse alternatives so that a person has multiple routes to consider [2]. Ustun *et al.* [3] explicitly constrain allowable feature changes for linear classifiers. These perspectives motivate separating the existence of a favorable feature vector from the feasibility of acting on it. The present comparisons use the repository's concrete adaptations of these ideas; they are not evaluations of every published variant.

B. CAUSAL RECOURSE AND MODEL UNCERTAINTY

Karimi *et al.* [4] establish intervention-based recourse: an action should be evaluated through a causal model rather than by assuming that all unchanged coordinates remain fixed. Their work is the closest conceptual foundation of CAC. Recourse under imperfect causal knowledge further addresses uncertainty about those mechanisms [5]. Our fitted-model rescore and synthetic audit are sensitivity analyses; they do not supply a posterior distribution over possible causal worlds.

Abduction, intervention, and prediction follow the structural causal perspective of Pearl [6]. A hand-specified graph combined with flexible regressors remains an assumed causal model. Conditional association tests can challenge some implications of that model but cannot identify all causal directions or individual effects. We consequently use the expression "usable under the specified SCM and constraints" throughout.

C. SCOPE OF THE EMPIRICAL COMPARISON

The primary saved experiment contains Random, a Wachter-style optimization, DiCE-random, a finite-grid point reference, and CAC. A minimal-cost causal intervention implementation, MINT, exists in the code and appears in the synthetic experiments. It is absent from the saved observational headline and search-seed comparison. We therefore make no observational claim that CAC improves on MINT or on the strongest available causal optimizer.

III. PROBLEM FORMULATION AND EVALUATION ENDPOINTS

Let $x \in \mathcal{X}$ be an encoded feature vector, f a fixed binary classifier, and $p_+(x)$ its favorable-class probability. The favorable decision is $f(x) = 1$ at threshold $\tau = 0.5$. We explain only test records for which $f(x) = 0$. An explanation comprises a proposed state z and a declared action $a = (S, v_S)$, where S is a feature subset and v_S gives its requested values. The classifier receives features only; an outcome node in an SCM graph is not an input coordinate.

A. REPLAY OF THE DECLARED ACTION

For a specified SCM, the action is replayed as

$$\widetilde{x} = \mathcal{T}_{\mathcal{M}}(x, a) = \mathcal{M}_{\mathrm{do}(X_S = v_S)}(u_x), \tag{1}$$

where u_x denotes the factual exogenous state inferred by abduction. The implementation propagates descendants in topological order and applies the feature encodings, bounds, and discrete rounding used by its saved models.

The audit uses the explicitly declared feature set, stored as changed_features. It does not infer extra interventions from every coordinate that differs between x and z. This matters for methods such as CAC, which report consequences as well as requested actions. If S is empty, replay returns the factual record. An unexplained favorable proposal therefore receives no credit simply because its state differs from x.

B. FOUR MEASUREMENTS AND ONE PRIMARY ENDPOINT

For each suggestion, proposed-state validity and replay success are

$$V(x, z) = \mathbf{1}\{p_+(z) \geq \tau\}, \tag{2}$$
$$R(x, a) = \mathbf{1}\{p_+(\widetilde{x}) \geq \tau\}. \tag{3}$$

Actionability, $A(x, a)$, requires admissible features and directions, at most 3 effective interventions, and normalized effort at most one. Plausibility is reported separately and is not an actionability gate.

To measure agreement between the promised state and the replayed state, let s_j be feature j's encoded range, lower bounded away from zero:

$$\Delta(x, z, a) = \max_j \frac{|z_j - \widetilde{x}_j|}{s_j}, \tag{4}$$
$$F(x, z, a) = V(x, z)\,\mathbf{1}\{\Delta(x, z, a) \leq \epsilon\}, \quad \epsilon = 0.1. \tag{5}$$

This is a range-normalized tolerance, not a standard-deviation tolerance. For categorical coordinates, the numerical gap is an encoding-dependent diagnostic and should not be interpreted as an intrinsic semantic distance.

The primary endpoint is

$$U(x, z, a) = R(x, a)A(x, a). \tag{6}$$

Neither V nor F gates U. A suggestion may fail at its named state but succeed after its consequences are propagated; a favorable replay can also remain usable while poorly describing its resulting state.

C. AGGREGATION AND DENOMINATORS

For applicant i and method m, with up to $k = 5$ suggestions, define

$$U_{im} = \max_{\ell \leq k} \big(R_{im\ell} A_{im\ell}\big), \qquad \widehat{U}_m = \frac{1}{n} \sum_{i=1}^{n} U_{im}. \tag{7}$$

The same suggestion must satisfy both conditions. Taking the maximum of replay success and the maximum of feasibility separately could credit different plans and would produce an incorrect endpoint. A method returning no suggestion contributes a failure for that applicant; the denominator remains the full query set. The other applicant-level columns use their own per-suggestion predicate, so they need not form a nested funnel.

IV. IMPLEMENTATION OF CAUSAL ACTIONABLE COUNTERFACTUALS

A. STRUCTURAL EQUATIONS

The repository specifies a directed acyclic graph for each dataset. Continuous nodes use fitted conditional means with additive factual residuals; discrete nodes use fitted conditional class probabilities with a fixed factual noise coupling. Histogram gradient boosting supplies the conditional estimators. Only descendants of intervened nodes are recomputed, and factual noise is reused across candidate actions for the same applicant. These choices provide a reproducible counterfactual coupling, not an identified individual mechanism.

B. ACTION EFFORT AND FEASIBILITY

Each feature has an effort function appropriate to its declared representation: linear or piecewise effort for numeric changes, ladder costs for ordinal levels, and entry costs for categorical switches. The action effort is

$$C(x,a) = \frac{\sum_{j\in S} w_j c_j(x_j, v_j)}{B}, \tag{8}$$

where the dataset-specific B sums the largest weighted maximum costs among the allowed number of actionable features. This normalizer reflects the configured action space. It does not measure an applicant's actual resources.

Costs and feasibility apply to requested interventions. A descendant that changes through propagation is not charged as an additional action. The audit nevertheless records side effects, including sensitive-feature changes. Hard restrictions on non-manipulable features and monotone directions are applied independently of the soft feasibility objective.

C. SEARCH OBJECTIVE

Let $q = \tau + 0.05$ for the favorable class, capped at one. Candidate actions are scored by

$$J(a;x) = \lambda_v \frac{[q - p_+(\mathcal{T}_{\mathcal{M}}(x,a))]_+}{q} + \lambda_c C(x,a) + \lambda_f H(x,a) + \lambda_p P(\mathcal{T}_{\mathcal{M}}(x,a)), \tag{9}$$

with weights $(\lambda_v, \lambda_c, \lambda_f, \lambda_p) = (1, 0.6, 0.3, 0.2)$. H combines the normalized inadmissibility penalty and sensitive-feature side effects. The hard legalization step typically makes the former zero. P is a range-normalized 5-nearest-neighbor distance to favorable-class training records, divided by a typical within-class reference distance. This plausibility penalty influences selection but is not a proof of distributional support.

Algorithm 1 Intervention-space search used by CAC

Require: Rejected x, classifier p_+, SCM $\mathcal{M}$, legal grids, k

Abduct factual noise u_x; initialize legal action population
Initialize archive of favorable candidate states
for each generation until iteration or patience limit **do**
  for each candidate action $a = (S, v_S)$ **do**
    Replay a using u_x to obtain $\widetilde{x}$
    Score $p_+(\widetilde{x})$ and objective (9)
    Archive favorable states with their declared actions
  end for
  Retain elites; crossover, mutate, and legalize children
end for
Select up to k distinct proposals using the stored objectives
return states, declared actions, effort and query metadata

D. POPULATION SEARCH AND REPORTING

Candidate values come from feature-specific grids built from training values or quantiles, with 12 quantile locations for numeric grids. A population of 60 candidates mixes singleton interventions with random multi-feature actions. Each generation retains the best 50%, applies uniform crossover, jitters selected values, and mutates the action subset. Legalization snaps values back to admissible grids and caps the intervention count. Search ends after at most 200 generations or 20 generations without a sufficient objective improvement.

The archive retains classifier-accepted candidates for later selection. If insufficient accepted candidates exist, the method can return best available non-flipping proposals; the external audit still scores them as failures. Algorithm 1 summarizes the search. It gives no global optimality or completeness guarantee. Batched classifier calls reduce overhead, while query counts charge scored candidate rows rather than treating an entire batch as one evaluation.

V. EXPERIMENTAL PROTOCOL AND EVIDENCE VERIFICATION

A. DATASETS AND PREDICTIVE MODELS

Table 1 reports the actual saved split sizes, feature counts, and predictive metrics for the primary XGBoost models [7]. Adult [8] predicts income above the dataset threshold; German [9] and the HELOC benchmark from FICO's challenge [10] use the favorable credit label. Diabetes [11] uses the inverted label no_diabetes=1-Outcome. Its feature interventions are a computational stress test and must not be interpreted as medical treatment. The saved model metadata marks all four observational datasets as non-synthetic. The two synthetic worlds are reported separately.

The query selector takes the first at most configured number of rejected test records in index order. It does not sample applicants afresh for each method. Each method receives the same records and returns up to 5 suggestions. The canonical search seed is 42. Search-seed repetitions keep the predictive model, training split, cost model, and SCM fixed.

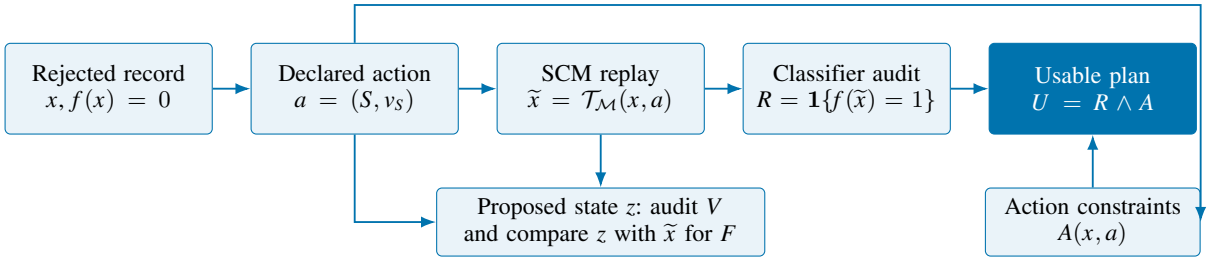


FIGURE 1. The evaluation separates a requested action from the state a method names and the state the SCM produces. The usability gate combines replay success and feasibility on one suggestion. Proposed-state validity and state fidelity are additional measurements. The SCM is an explicit assumption.

TABLE 1. Saved observational datasets and primary classifier performance. "Queries" counts rejected test applicants selected for recourse; d counts input features. Accuracy and area under the ROC curve (AUC) are fractions. These predictive estimates inherit the protocol limitations in Section VIII.

Dataset	Train	Test	d	Queries	Acc.	AUC
Adult	16000	4000	9	100	0.8485	0.9182
German	800	200	15	51	0.7750	0.7846
HELOC	7896	1975	12	100	0.7205	0.7880
Diabetes	614	154	8	43	0.7727	0.8489

B. BASELINE IMPLEMENTATIONS AND ACCESS ASSUMPTIONS

Random samples candidate feature changes. The Wachter-style implementation uses finite-difference L-BFGS-B with restarts, and DiCE uses the library's random method. The finite-grid reference performs cost-ordered direct point search with an evaluation cap. It is called "oracle" in stored artifacts, but its feasible set differs from causal intervention search. Its cost is therefore neither a universal causal lower bound nor an optimality certificate for CAC.

All primary methods operate through prediction interfaces. The baselines receive immutability restrictions, but their optimization procedures and query budgets are different. Finite differences can be a poor fit to a piecewise-constant tree ensemble. Query counts are consequently reported as descriptive computational costs, not as evidence from an equal-budget contest. The strongest causal baseline available in the source, MINT, optimizes action cost through the SCM but has only synthetic results in the saved comparisons used here.

C. VERIFICATION AGAINST SAVED ARTIFACTS

The new evidence builder reconstructs the main table from counterfactuals.csv, faithfulness_per_cf.csv, and feasibility_per_cf.csv. It verifies unique suggestion keys, matching applicant sets, audit coverage, and agreement with all reported headline fields. It additionally reloads each dataset's saved counterfactual objects and recomputes their metrics, declared-action replay, feasibility, and headline summaries against the currently stored models.

Saved cost ratios differ from ratios recomputed under the current default filtering convention. Rather than mixing these conventions, this paper reports absolute normalized action effort. The discrepancy is retained in the verification outputs. Predictive metrics are cross-checked against models/metrics.json; the report includes SHA-256 hashes of the inputs and relevant code. This verifies the saved evidence and current replay; it does not claim that every expensive search experiment was rerun.

A fresh local execution of the project's test suite produced 386 passes, 1 skip, and no failures or setup errors. The skipped test concerns dropping a sub-plan when the generated action has only one feature. Unit tests establish software properties on their fixtures; they do not validate a causal graph or an empirical superiority claim.

D. APPLICANT-LEVEL STATISTICAL ANALYSIS

Each baseline is paired with CAC on the identical applicant at the canonical seed. We use the exact two-sided McNemar test and apply Holm correction at familywise level 0.05 within each dataset. The paired sample size is the number of distinct applicants, never applicants multiplied by search seeds.

For effect sizes, we first average each applicant's paired usability difference over shared available seeds, then bootstrap whole applicants with replacement using 10,000 resamples at seed 42. The interval therefore describes variation across queried applicants conditional on the saved split and models. Its estimand differs from the single-seed binary test. Observed per-seed means and ranges are reported separately, with the actual number of available seed runs. Missing runs are not imputed and recorded generation exceptions remain failures.

VI. RESULTS

A. PRIMARY COMPARISON: REPLAY AND FEASIBILITY EXPOSE DIFFERENT FAILURES

Table 2 reports the reconstructed canonical-seed comparison. CAC attains usable recourse for all 294 queried applicants under the shared SCM and constraints. This outcome does not imply uniform superiority: Random and DiCE-random tie it on Adult and German, and several methods tie it on diabetes.

The minimum saved usable rate in this table is 18.0%, not the lower percentage stated in an older manuscript.

HELOC exhibits the clearest separation. Random has 96.0% proposed-state validity but 78.0% usable recourse; DiCE-random has 100.0% and 76.0%, respectively. The Wachter-style method has 85.0% proposed-state validity, 72.0% replay success, and 35.0% actionability, with only 18.0% of applicants receiving a plan that passes both latter tests. These rates cannot be multiplied to recover usability because they are applicant-level marginal events over multiple suggestions.

B. EFFORT AND COMPUTATIONAL COST DO NOT SUPPORT A UNIVERSAL RANKING

CAC's mean normalized effort is 0.1807 on Adult, 0.0091 on German, 0.0308 on HELOC, and 0.0275 on diabetes. The Adult Random baseline achieves the same applicant coverage at lower reported effort. In contrast, CAC on HELOC has higher usable coverage than Random and lower conditional effort. These comparisons describe the saved outputs: effort averages condition on each method's own successful applicants and do not constitute a paired common-success cost comparison.

The recorded query counts also vary substantially across methods and datasets. A low count can accompany a less reliable proposal, and an expensive search can still fail the action audit. Neither query count nor the objective value alone summarizes the coverage–effort tradeoff. A future equal-budget study should compare success curves and paired costs on a prespecified common-success set.

C. SEARCH REPETITIONS REVEAL MISSING COVERAGE AND RECORDED FAILURES

The saved study intends 5 seeds: 42,0,1,2,3. Actual coverage differs. Adult has only 4 runs for Wachter, DiCE-random, and the grid reference; their final intended seed is absent. Table 3 reports means and observed ranges over the available runs, without presenting a complete five-seed design for those methods.

CAC remains at full usability across the saved Adult, German, and HELOC seeds, but its diabetes mean is 94.4%, with a minimum of 72.1%. That minimum corresponds to a run recording 12 generation exceptions; the original run metadata does not establish their cause. They are retained as failures rather than removed from the denominator. Adult DiCE-random has a 75.0% mean and a 0.0% minimum across its available runs, even though no generation exceptions are recorded for the zero-usable run. This is evidence of a search or run anomaly, not a justified reason to erase the observation.

D. CORRECTED INFERENCE USES APPLICANTS AS INDEPENDENT UNITS

Table 4 replaces the saved pooled summary. At the canonical seed, CAC's advantage over every HELOC baseline survives the within-dataset Holm correction. The German Wachter comparison and Adult grid-reference comparison also survive correction. The remaining canonical comparisons do not establish a difference at the specified family level.

The diabetes comparisons illustrate the distinction between estimands: canonical CAC succeeds for every applicant, while the failed search-seed run reduces its across-seed average. Similarly, Adult DiCE-random ties CAC at the canonical seed while its anomalous saved run changes the across-seed effect. A bootstrap interval for the mean across available seeds must not be described as the interval associated with the canonical binary McNemar hypothesis. Neither analysis includes variation over new training splits, learned graphs, or applicant populations.

E. ABLATION ISOLATES PROPAGATION AND THE COST OBJECTIVE

Table 5 compares the full method with removing propagation and with setting the cost coefficient to zero. The no-propagation variant uses the same action grids, legalization, and search operators but scores directly edited states. Its usability falls on Adult and HELOC and remains at full coverage on German and diabetes. Thus the saved ablation supports a dataset-dependent benefit of propagation, not the claim that removing it always destroys usable recourse.

Removing the effort term raises mean effort in every dataset in the saved ablation. This result is consistent with the objective influencing which successful plans are preferred. Removing the soft feasibility coefficient leaves the hard action restrictions in place and gives the same recorded outputs as the full method. Consequently, that ablation cannot show that feasibility constraints are unnecessary.

VII. SENSITIVITY TO THE CAUSAL MODEL

A. REFITTED EQUATIONS: A SENSITIVITY CHECK WITH OVERLAPPING DATA

Table 6 re-audits the same plans under a refitted FCM. CAC usability changes to 93.0% on Adult and 97.0% on HELOC, while remaining at full coverage on German and diabetes. Suggestion-level survival falls further than applicant-level coverage because an applicant needs only one surviving feasible plan.

The saved artifact contains only the rescore arm. Its original FCM was trained on the full training split; the alternative FCM uses one half of that split. These fitting sets overlap. Calling this result a disjoint or independent causal validation would therefore overstate its design. The code also implements a separate cross-fit arm that would generate on one half and audit on the other, but no such results are present in the saved table. We do not substitute the code's capability for experimental evidence that was not saved.

B. KNOWN SYNTHETIC MECHANISMS

The synthetic Reach and Block worlds provide analytic structural equations for a separate replay auditor. Both query 100 rejected applicants. Under the exact equations, CAC and MINT both provide full applicant-level coverage in Reach and 40.0% in Block (Table 7). MINT has lower mean action

TABLE 2. Verified main-study results at search seed 42. All rates are percentages of the full applicant query set: proposed-state validity (*V*), replay success (*R*), actionability (*A*), state fidelity (*F*), and usability (*U*). Each column credits any passing suggestion, but *U* requires *R* and *A* on the same suggestion. Cost is mean normalized effort of the cheapest usable plan per successful applicant; its denominator varies by method. Queries are mean scored classifier-input rows per applicant, excluding common query-set selection and subsequent external auditing. Blue rows identify CAC.

Dataset	Method	n	V	R	A	F	U	Cost	Queries
Adult	Random	100	100.0	100.0	100.0	100.0	100.0	0.0439	1,006.0
Adult	Wachter	100	100.0	100.0	99.0	100.0	99.0	0.1113	17,439.0
Adult	DiCE-random	100	100.0	100.0	100.0	100.0	100.0	0.0745	2,022.6
Adult	Grid reference	100	100.0	87.0	100.0	99.0	87.0	0.1225	3,984.4
Adult	CAC	100	100.0	100.0	100.0	100.0	100.0	0.1807	1,032.6
German	Random	51	100.0	100.0	100.0	92.2	100.0	0.0025	1,006.0
German	Wachter	51	100.0	98.0	74.5	82.4	70.6	0.0174	52,953.4
German	DiCE-random	51	100.0	100.0	100.0	96.1	100.0	0.0082	2,058.6
German	Grid reference	51	100.0	100.0	100.0	94.1	100.0	0.0012	95.1
German	CAC	51	100.0	100.0	100.0	100.0	100.0	0.0091	1,216.0
HELOC	Random	100	96.0	90.0	91.0	69.0	78.0	0.0846	1,005.7
HELOC	Wachter	100	85.0	72.0	35.0	48.0	18.0	0.0498	37,379.8
HELOC	DiCE-random	100	100.0	78.0	91.0	64.0	76.0	0.0934	2,609.5
HELOC	Grid reference	100	100.0	83.0	100.0	82.0	83.0	0.0255	7,714.2
HELOC	CAC	100	100.0	100.0	100.0	100.0	100.0	0.0308	1,312.8
Diabetes	Random	43	100.0	97.7	95.3	86.0	93.0	0.0362	1,006.0
Diabetes	Wachter	43	100.0	100.0	100.0	93.0	100.0	0.0645	7,314.3
Diabetes	DiCE-random	43	100.0	100.0	100.0	100.0	100.0	0.0549	2,025.2
Diabetes	Grid reference	43	100.0	100.0	100.0	97.7	100.0	0.0241	174.0
Diabetes	CAC	43	100.0	100.0	100.0	100.0	100.0	0.0275	835.5

TABLE 3. Usability over the available search-seed runs: mean [minimum, maximum] in percent, followed by the number of saved runs in parentheses. Models and applicants are fixed across seeds. These are observed optimizer summaries, not confidence intervals for performance on new applicants.

Method	Adult	German	HELOC	Diabetes
Random	100.0 [100.0, 100.0] (5)	100.0 [100.0, 100.0] (5)	82.6 [78.0, 87.0] (5)	95.3 [93.0, 97.7] (5)
Wachter	98.8 [98.0, 99.0] (4)	77.3 [70.6, 82.4] (5)	19.4 [18.0, 21.0] (5)	94.4 [88.4, 100.0] (5)
DiCE-random	75.0 [0.0, 100.0] (4)	100.0 [100.0, 100.0] (5)	75.8 [73.0, 80.0] (5)	99.5 [97.7, 100.0] (5)
Grid reference	87.0 [87.0, 87.0] (4)	100.0 [100.0, 100.0] (5)	83.0 [83.0, 83.0] (5)	100.0 [100.0, 100.0] (5)
CAC	100.0 [100.0, 100.0] (5)	100.0 [100.0, 100.0] (5)	100.0 [100.0, 100.0] (5)	94.4 [72.1, 100.0] (5)

effort in these saved comparisons; CAC's additional objective terms do not establish a general cost advantage.

The Reach result also shows why full applicant coverage does not mean every suggestion survives: CAC's suggestion-level truth survival is 95.6%. In Block, the saved intrinsic reachability diagnostic marks 42 of 100 queries as blocked by the analytic data-generating mechanism. The lower observed usable coverage is an outcome of the fixed learned classifier and finite search as well as the mechanism. The unserved fraction therefore must not be equated with that intrinsic blocked count or described as an exact impossibility proof for every unsuccessful applicant.

C. OBSERVATIONAL GRAPH DIAGNOSTICS

Table 8 summarizes the saved graph checks. The graphs contain the outcome as well as feature nodes, so their node counts differ from the classifier input dimension in Table 1. Edge-support and conditional-independence tests provide observational diagnostics of the chosen graphs. German and HELOC have substantial unsupported or unresolved structure, which limits an interpretation based solely on internal replay success.

VIII. LIMITATIONS AND RESPONSIBLE INTERPRETATION

A. PREDICTIVE PROTOCOL AND DATA PROVENANCE

The saved predictive metrics are internally consistent with their model reports, but the training protocol is not a fully isolated test evaluation. The model-training code checks test AUC to decide whether to trigger a hyperparameter search. Although the final tuned/default choice uses training cross-validation, the decision to search has already consulted the test set. In addition, HELOC and diabetes median imputation occurs before the split, and the diabetes cleanup uses class-dependent patterns to identify a redistributed source's suspected label-dependent fill values. These steps limit claims about out-of-sample predictive performance.

A stronger confirmatory experiment should split before estimating any preprocessing quantities, retain an original

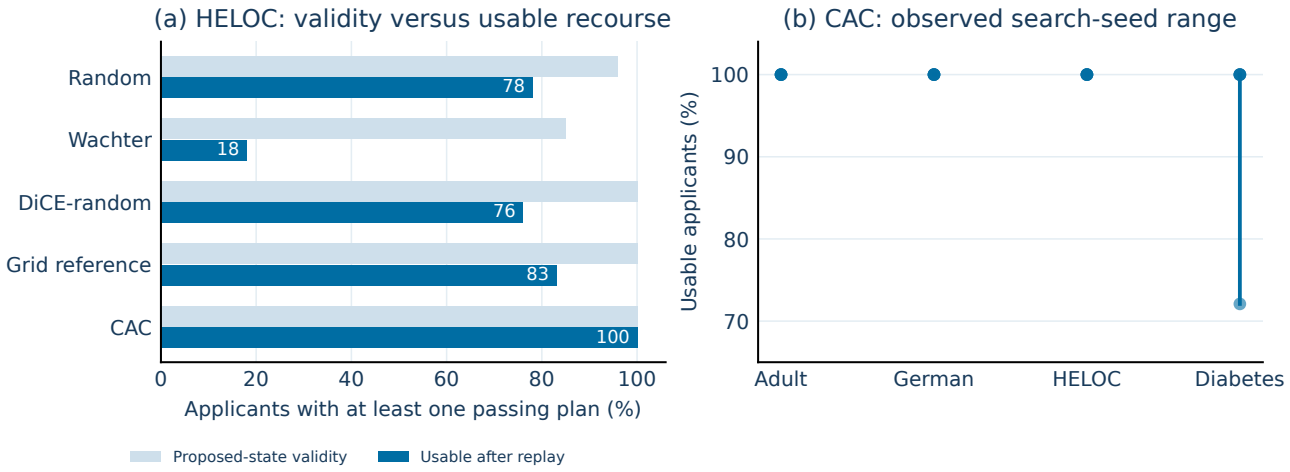


FIGURE 2. **Plots generated from the verified tables. (a) HELOC illustrates why proposed-state validity is insufficient to describe usable recourse. (b) The observed CAC search-seed range retains the diabetes run with generation failures. Coincident markers at full success do not represent zero population uncertainty.**

TABLE 4. **Recomputed inference from saved applicant-level records. n is the number of distinct paired applicants; "Seeds" is the number shared for effect-size estimation. b/c counts canonical-seed applicants for whom only CAC/only the baseline supplies usable recourse. Gap and its percentile interval average paired differences across shared seeds and resample applicants, in percentage points (pp). McNemar p values use only seed 42 and are Holm-adjusted within dataset. These two estimands need not agree in sign or apparent significance.**

Dataset	Baseline	n	Seeds	b/c	Gap (pp)	95% cluster CI	Holm p
Adult	Random	100	5	0/0	+0.0	[+0.0, +0.0]	1.000
Adult	Wachter	100	4	1/0	+1.2	[+0.0, +3.0]	1.000
Adult	DiCE-random	100	4	0/0	+25.0	[+25.0, +25.0]	1.000
Adult	Grid reference	100	4	13/0	+13.0	[+7.0, +20.0]	9.77×10^{-4}
German	Random	51	5	0/0	+0.0	[+0.0, +0.0]	1.000
German	Wachter	51	5	15/0	+22.7	[+14.1, +32.2]	2.44×10^{-4}
German	DiCE-random	51	5	0/0	+0.0	[+0.0, +0.0]	1.000
German	Grid reference	51	5	0/0	+0.0	[+0.0, +0.0]	1.000
HELOC	Random	100	5	22/0	+17.4	[+12.4, +22.4]	9.54×10^{-7}
HELOC	Wachter	100	5	82/0	+80.6	[+73.4, +87.4]	1.65×10^{-24}
HELOC	DiCE-random	100	5	24/0	+24.2	[+17.4, +31.4]	3.58×10^{-7}
HELOC	Grid reference	100	5	17/0	+17.0	[+10.0, +25.0]	1.53×10^{-5}
Diabetes	Random	43	5	3/0	-0.9	[-6.5, +5.6]	1.000
Diabetes	Wachter	43	5	0/0	+0.0	[-3.7, +4.2]	1.000
Diabetes	DiCE-random	43	5	0/0	-5.1	[-7.9, -2.3]	1.000
Diabetes	Grid reference	43	5	0/0	-5.6	[-8.4, -2.8]	1.000

missingness mask and verified source, and make all tuning decisions using training/validation data. The current paper preserves the measured results and discloses their protocol rather than asserting that an artifact consistency check repairs these design limitations.

B. CAUSAL UNCERTAINTY AND OPERATIONAL MEANING

The learned equations, discrete noise coupling, graph directions, cost levels, and legal interventions are assumptions. Passing their audit means that a computational plan is consistent with those assumptions. It does not demonstrate that a person can obtain the predicted income, credit status, or physiological change. Costs do not measure individualized money, time, access, or burden, and the models do not capture dynamic adaptation or uncertainty in treatment response.

The computational diabetes benchmark is particularly constrained: changing a measurement in an SCM is not evidence of an effective clinical action. No result in this paper supports treatment recommendations. Likewise, omitting protected features from direct interventions does not prove counterfactual fairness or absence of harmful downstream effects.

C. COMPARISON BREADTH AND UNCERTAINTY

This study uses fixed saved models and a small deterministic subset of their rejected records. Search-seed variation does not represent variation over population samples or training splits. Incomplete Adult repetitions, the diabetes exceptions, and the Adult DiCE anomaly prevent a clean claim of uniform search stability. Unequal compute budgets and the lack of an

TABLE 5. Ablation at the canonical seed. *U* is applicant-level usability in percent. The cost columns are mean effort of the cheapest usable plan; "Ratio" is the quotient of these means for the no-cost-term variant and the full method, not the repository's median Cost/CAC statistic.

Dataset	*U* full	*U* no prop.	Cost full	No cost term	Ratio
Adult	100.0	95.0	0.1807	0.2467	1.37
German	100.0	100.0	0.0091	0.0940	10.37
HELOC	100.0	96.0	0.0308	0.0519	1.68
Diabetes	100.0	100.0	0.0275	0.0346	1.26

TABLE 6. CAC under the saved refitted-equation rescore. *U* is a percentage of all queried applicants. *Survival is the percentage of individually valid proposed suggestions that remain accepted after replay, a different denominator. The refitted auditor reuses a subset of the generator's training rows.

Dataset	*n*	*U* own	*U* refit	Survival*
Adult	100	100.0	93.0	73.0
German	51	100.0	100.0	97.3
HELOC	100	100.0	97.0	79.4
Diabetes	43	100.0	100.0	86.5

TABLE 7. Saved synthetic-world audit. *U* is usable-applicant percentage under the fitted generator SCM (own) and the analytic SCM (truth). *Survival is conditional on individually valid proposed suggestions, not all applicants. Cost is mean effort of the cheapest truth-usable plan per successful applicant. These figures are checked against the saved summary; individual synthetic plans are not persisted for fresh replay.

World	Method	*n*	*U* own	*U* truth	Surv.*	Cost truth
Reach	CAC	100	100.0	100.0	95.6	0.0529
Reach	MINT	100	100.0	100.0	91.6	0.0202
Block	CAC	100	40.0	40.0	93.0	0.0706
Block	MINT	100	40.0	40.0	95.0	0.0469

TABLE 8. Saved SCM diagnostics. "Actions" counts manipulable feature nodes; "Supported" is supported asserted edges over all asserted edges; "CI upheld" is the saved conditional-independence test count. These are diagnostics of observational implications, not proofs of causal direction.

Dataset	Nodes	Edges	Actions	Supported	CI upheld
Adult	10	25	7	17/25	19/35
German	16	37	11	18/37	88/138
HELOC	13	28	8	18/28	40/75
Diabetes	9	17	3	16/17	27/27

observational MINT comparison limit algorithmic rankings. The synthetic audit adds a known mechanism but remains a stylized test rather than a substitute for a domain-validated causal model.

The repository includes a user-study protocol and synthetic responses generated to exercise an analysis pipeline. Those responses are not human observations and are excluded from the empirical evidence. We report no measured improvement in trust, comprehension, human outcomes, or actual ability to execute a recommendation.

IX. CONCLUSION

The saved experiments show that predictive validity, replay success, actionability, and fidelity answer different questions. CAC supplies usable plans for every applicant in the canonical shared-model comparison, but its ties with baselines, nonuniform seed behavior, and sensitivity to fitted equations delimit that result. An intervention representation makes causal assumptions explicit; it cannot make those assumptions true.

Evaluation should therefore retain declared actions, audit their replay, state the usability conjunction, keep failed applicants in denominators, and distinguish search variation from applicant-level inference. The verified tables provide an auditable account of this implementation and a concrete basis for a subsequent equal-budget, disjoint-model, provenance-controlled study.

DATA AND CODE AVAILABILITY

The accompanying project contains the method implementations, experiment scripts, saved results, and tests. The new manuscript's values, tables, and vector plots are generated by paper/build_access_evidence.py. Its verification directory contains the fresh test report, reconstructed headline, fresh saved-plan replay, corrected applicant-level statistics, discrepancy reports, and source hashes. Rebuilding requires the corresponding saved artifacts; no public repository URL or archival identifier has been assigned in this draft.

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